

Inverse Trigonometric Functions.

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 120^\circ = \frac{\sqrt{3}}{2}$$

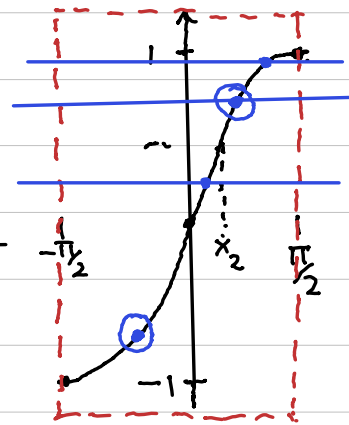
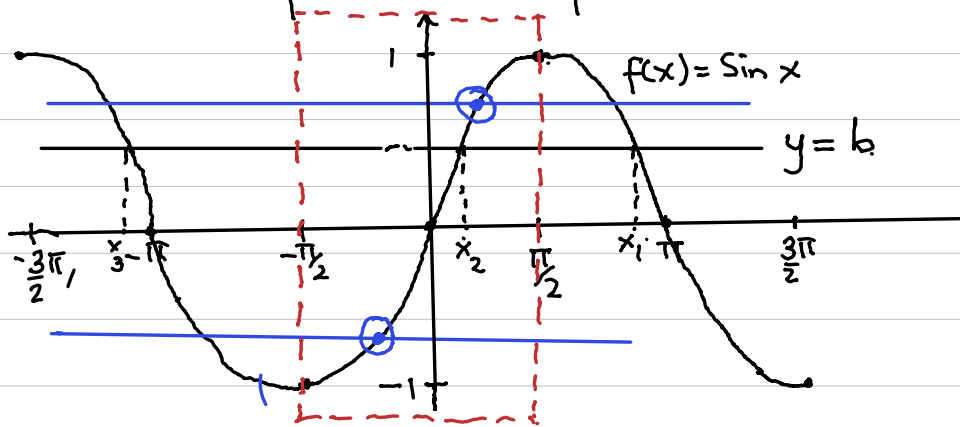
$$\sin 420^\circ = \frac{\sqrt{3}}{2}$$

✓ Trig. fns are not 1-1, hence do not have inverse functions.

✓ We can however restrict their domains so that they become 1-1.

Example.

Consider the function $f(x) = \sin x$



- Quickest way to check 1-1ness is by using the horizontal line test. $f(x) = \sin x$ fails the test.
- However, the inverse fn of the restricted sine fn exists and is denoted by $\sin^{-1}$ or $\arcsin$.

Definition

~~$\sin$~~

$$\begin{array}{ccc} \sin^{-1} x = y & \iff & \sin y = x \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \\ \parallel & & \parallel \\ \sin^{-1}(\sin y) = y & & \sin(\sin^{-1} x) = x \end{array}$$

✓ In the same way, we can define the rest of the ^{inv.} trig. functions as follows.

$$\begin{array}{lcl} \cos^{-1} x = y & \iff & \cos y = x \quad \forall y \in [0, \pi] \\ \tan^{-1} x = y & \iff & \tan y = x \quad \forall y \in (-\frac{\pi}{2}, \frac{\pi}{2}) \quad 0 \leq y \leq \pi \end{array}$$

✓ The "cancellation" equations become

$$\begin{array}{ll} \sin^{-1}(\sin x) = x & \forall x \in [-\pi/2, \pi/2] \\ \sin(\sin^{-1} x) = x & \forall x \in [-1, 1] \end{array}$$

Example

Evaluate: a) $\sin^{-1}(1/2) = \frac{\pi}{6}$

b) $\cos(\sin^{-1}(1/2)) = \sqrt{3}/2$

c) $\tan(\sin^{-1} x)$.

Solⁿ ©

(Let) $\theta = \sin^{-1} x \iff \sin \theta = x \quad \forall x \in [-1, 1] \text{ and } \forall \theta \in [-\pi/2, \pi/2]$

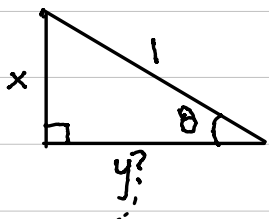
✓ Thus, we ~~need~~^{want} to find $\tan \theta$.

Method 1

We have it that $\sin \theta = \frac{x}{1}$.

$$\sin \theta = \frac{\text{opp.}}{\text{Hyp}}$$

$$\tan \theta = \frac{\text{opp.}}{\text{Adj}}$$



By the Pythagoras thm, $y = \sqrt{1-x^2}$.
Hence,

$$\tan \theta = \frac{x}{\sqrt{1-x^2}} = \tan(\sin^{-1} x).$$

Method 2 (Pythagoras thm in disguise)

- With $\sin \theta = x$, we can find $\cos \theta$.

- We know that

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \implies \cos \theta = \sqrt{1 - \sin^2 \theta} \\ &= \sqrt{1 - x^2} \end{aligned}$$

$$\text{- And } \tan \theta = \tan(\sin^{-1} x) = \frac{\sin \theta}{\cos \theta} = \frac{x}{\sqrt{1-x^2}}.$$

Exercise

Remember the definition.

1. Simplify

a) $\cos(\tan^{-1}x)$

b) $\tan(\cos^{-1}x)$

c) $\sin(2 \cos^{-1}x)$

2. Prove that $\underbrace{\sin^{-1}x}_{\theta_1} + \underbrace{\cos^{-1}x}_{\theta_2} = \pi/2$

Hint:

Use $\sin(\theta_1 + \theta_2) = \sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_1$.

Derivatives of Inverse trig. fns.

✓ Using the chain rule, we can have the following.

* $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$ and $\frac{d}{dx}(\sin^{-1}u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$ $u=g(x)$

$$\frac{d}{dx}(\cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}} \quad \forall x \neq \pm 1$$

$$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\csc^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}} \quad \forall x \neq \pm 1, 0$$

Check!!

$$\frac{d}{dx}(\sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}} \quad \forall x \neq \pm 1, 0$$

(For) *
 Let $y = \sin^{-1}x$. $\frac{dy}{dx}$ $x = \sin y$ $\leftarrow y$

$$\Rightarrow \frac{d(\sin y)}{dx} = \frac{dx}{dx}$$

$$\cos y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

Since $x = \sin y$

$\forall x \neq \pm 1$.

Example.

Find the derivative of
a) $\cos(\sin^{-1}x)$

Solⁿ

Let $y = \cos(\sin^{-1}x)$ and $u = \sin^{-1}x$. Then $y = \cos u$.

Hence,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= -\sin u \times \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{-\sin u}{\sqrt{1-x^2}}$$

$$= \frac{-\sin(\sin^{-1}x)}{\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}.$$

b) $y = x \sec^{-1}(\sqrt{x})$ Exercise:)